

26 GIGAWATT

THE GLOBAL WINDPOWER CAPACITY IN 2007

The IPCC stressed the need to increase the proportion of electricity generated by renewable sources, such as wind power, to mitigate climate change.



Skull of Selam

This well-preserved skull is that of a human ancestor nicknamed Lucy's Child, who would have walked upright on two legs.

important find: a partial skeleton of a **bipedal human ancestor** who lived and died around **3.3 million years ago**. The fossilized remains were of a young female about three years of age, and had been discovered in 2000, in Dikika, Ethiopia—close to where the skeleton nicknamed Lucy was found (see 1974). The individual, from the same species as Lucy (*Australopithicus afarensis*), earned the nickname Lucy's Child—although she predates Lucy by around 120,000 years.

IN ITS FOURTH ASSESSMENT REPORT, the Intergovernmental Panel on Climate Change (IPCC) further refined and extended its analysis of **global climate change** presented in its earlier reports (see 1990, 1995, 2001), while reiterating the same conclusions with still more certainty: that human activities—particularly carbon dioxide emissions from burning fossil fuels—are producing a greenhouse effect (see pp.326–27), causing a **rise in average global temperature**.

While mobile phones were becoming commonplace in developed nations, most devices could do little more than make calls and send SMS (text)

messages. American company Apple Inc. **revolutionized the mobile phone industry** with the introduction of the iPhone. In addition to calling and texting, users could browse the web, download and use a huge range of apps (applications), listen to music, watch and record videos, and take pictures. The iPhone was an enormous success, and other manufacturers soon created smartphones with similar capabilities.

Stem cell research promises many benefits in medicine—for example, stem cells could regenerate brain cells to cure dementia. In an example of this kind of potential, researchers in

Apple iPhone

The iPhone from Apple introduced a convenient touch interface, with gestures such as swiping and pinching, to mobile phones.

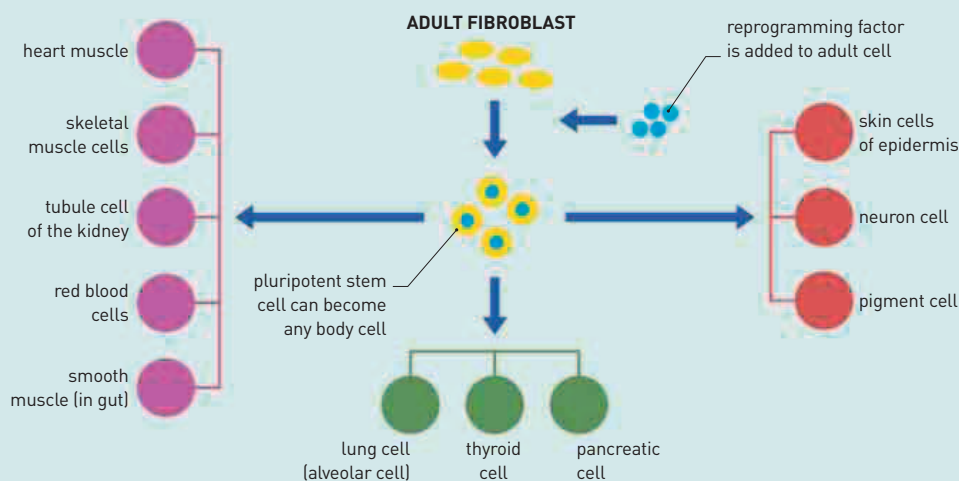
the UK **grew heart tissue** from bone marrow stem cells in 2007. One stumbling block in stem cell research was the fact that only embryos contain cells that can develop into any kind of cell. Many people found that idea objectionable, because the embryos were destroyed in the process. However, in 2007 two teams of scientists independently managed to **reprogram ordinary cells** (see panel, below) to make them into stem cells—a feat previously



achieved with cells taken from mice. Also this year, a team used DNA bases to **synthesize a copy of a bacterial chromosome**.

CELL PROGRAMMING

A major breakthrough in stem cell research involves cells called fibroblasts, which are found in skin and connective tissue. Fibroblasts are responsible for producing collagen and other proteins that repair the skin. Turning these cells into pluripotent stem cells, which have the potential to develop into any kind of cell, involves adding compounds that switch on certain genes. This causes the cells to revert to the state in which all cells begin.



October 6 Astronomers report the most rigorous test of Einstein's general theory of relativity, through **observations of a double pulsar**

October 30 Publication of the Stern Review, a forecast of the **economic consequences of global warming**

April 6 Final version of Climate Change 2007, the IPCC's **Fourth Assessment Report** is published

April 24 Discovery of a **potentially habitable exoplanet** and evidence of water vapor on another

April Human heart valve grown from human bone marrow stem cells

June 14 Paleontologists report finding fossils of a **giant birdlike dinosaur, Gigantoraptor erlianensis** in Inner Mongolia

June 29 Apple releases the iPhone—the **first successful smartphone**

October 6 American biologist Craig Venter announces his team has **synthesized a copy of a simple bacterium**

November 20 Two teams of scientists report they have **produced human stem cells** by reprogramming normal cells